Himanshu Bhenwal

Bachelor of Technology
Computer Science Engineering and DS and AI

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Github LinkedIn

Education

B.Tech in Computer Science and Engineering *2021-25*

Bhagwan Parshuram Institute Of Technology, New Delhi CGPA : 8.95/10

Experience

Machine Learning Engineer *January 2026 - Present*

PrudentBit, Noida, Uttar Pradesh

- Architected a secure document intelligence system for PII detection and redaction, supporting high-volume enterprise workflows.
- Implemented multilingual NER pipelines across structured and unstructured data sources, enabling processing of 10k+ documents per day.
- Improved entity detection performance by integrating transformer-based models with rule-based validation, increasing F1-score by 10%.

Researcher Intern *July 2025 - January 2026*

Lossfunk, Bengaluru, Karnataka

- Conducted research on model-based reinforcement learning to improve generalization and reduce sample complexity.
- Developed prototype world models for high-dimensional environments, reducing training steps required for policy convergence.
- Analyzed training stability techniques, leading to more consistent long-horizon predictions in simulated environments.

Machine Learning Engineer Intern *Jun 2024 - July 2025*

PrudentBit, Noida, Uttar Pradesh

- Built core modules of Immunefiles-PII, a production system for sensitive data detection using hybrid ML and rule-based approaches.
- Collaborated with senior leadership to deploy scalable pipelines handling real-world document processing workloads.
- Enhanced precision-recall tradeoff through iterative experimentation, reducing false positives in noisy datasets.

Personal Projects

Playing MsPacMan using RL and VQVAE based world models

End-to-end pipeline for learning compact latent policies on MsPacMan.

- Trained a Vector-Quantized VAE to compress raw game frames into a discrete latent space.
- Learned DQN agents directly on the learned latents, with and without PER, for Ms Pac-Man.
- Built value, world-model, and action-mapping networks to enable model-based planning.
- Provided utilities for dataset generation, mixed-precision training, and GPU/MPS support.
- Used Python, PyTorch and WandB for experiment tracking.

sushi-GGUF : A minimalist framework for GGUF quantization of SDXL models

Implemented quantization techniques to build a tool for GGUF quantizing Stable Diffusion models.

- Used Python for the main scripting.
- Provided options to quantize to various precision levels such as Q4KS, Q5KS, Q80.
- Used llama.cpp binary files for performing quantization for each quantization precision.

Paper Implementations

Github respository containing several research paper implementations.

- Used Python, PyTorch, WandB for experiment tracking.

Achievements, Open Source Contributions and Talks

Served as member of jury at the Datadive : The Ultimate Datathon

- Served as a judge at the annual flagship datahon organized by IEEE WIE BPIT.

Talks

- **July 2024** Transformers to Mamba, AI4Bharat
- **February 2025** An Introduction To Reasoning Capabilities in Language Models, IEEE BPIT

Conference Attendances

- **August 2024** Selected among 5 exceptional student volunteers in the IEEE Student Branches of New Delhi, India to attend the IEEE Region 10 SYWLC Congress 2024 at Tokyo, Japan held from 29th August to 1st September 2024.

Positions Of Responsibility

Chairperson at IEEE BPIT Student Branch *2021 - 2025*

- Mentored a community of 80+ students, fostering a culture of continuous learning and growth.
- Initiated impactful initiatives to promote Machine Learning awareness among peers

Co-creator and Lead Instructor of IEEE BPIT's SIG on ML *2022 - 2025*

- Teaching ML in semester long classes to students from different branches.
- Leading weekly paper reading sessions and discussions with fellow team members.
- Helping fellow junior students in their research projects in Computer Vision, NLP.

Technical Skills and Interests

Languages: C, C++, Python
Libraries: NumPy, Pandas, C++ STL, Transformers, LangChain, LlamaIndex
Dev Tools/Environments: VScode, Vim, JupyterLab, Git, Docker, GitHub Actions
Frameworks: TensorFlow, PyTorch, JAX, Keras, scikit-Learn, FastAPI, Django, Celery
Databases: PostgreSQL, Redis
Cloud Platforms: AWS (Lambda, EC2, S3, SageMaker)